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COMP4139

Machine Learning (MLE)

- Q&A

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Why do we normalise data?



- Benefit of data normalisation:
 - ✓ Improves the numerical stability of the machine learning model. (e.g. using gradient descend to optimise neural network, if different features are in different value range a fixed learning rate will likely overshoot or undershoot the optima for certain features).
 - ✓ Reduce the negative impact to distance-based algorithms (e.g. KNN, SVM).



What is curse of dimensionality?



Curse of Dimensionality

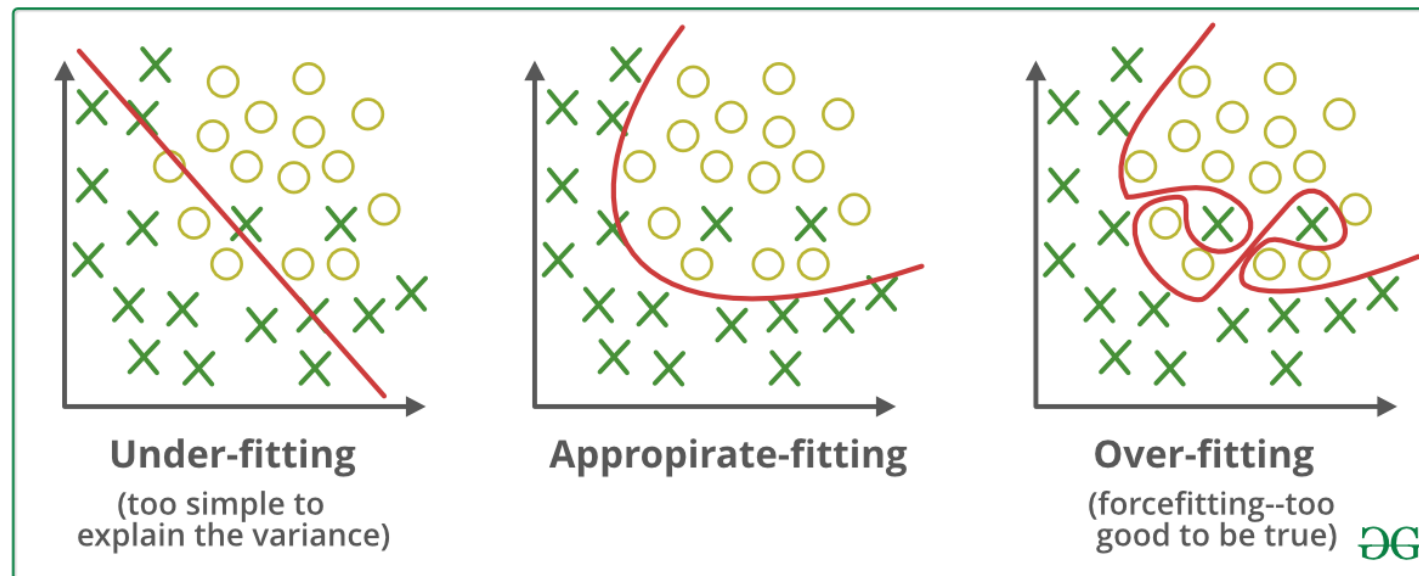
- The required sample numbers will need to increase exponentially with a linear increase of feature dimensions.
- What shall we do to avoid curse of dimensionality?
 - Increase the number of data samples.
 - Reduce the number of features (more feasible): use feature selection and dimensionality reduction methods.



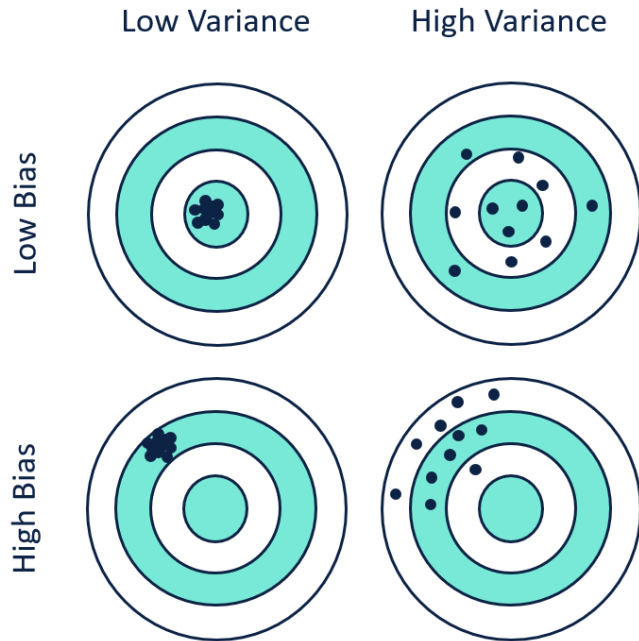
What is overfitting how do we avoid it?

Overfitting vs Underfitting

- Overfitting: a model learns to map the training data too well, which negatively impacts the performance of the model on new, unknown, data. **Poor generalisability.**
- Underfitting: if it neither can model the training nor the test data correctly. Underfitting is easy to detect in the training phase using evaluation metrics. **Poor performance.**



- **Bias** indicates the error between the approximated model to the ideal model.
- **Variance** is the amount that the estimate of the target function will change given different training data.





If I have two categorical variable,
which statistical test I can use to
see significant association?

Chi Square vs T-test vs ANOVA (Filter Method)

- Univariate feature selection (assuming features are independent to each other)
- A chi-square tests the independence of predictor and outcome event, suitable for **categorical** features in **categorical** outcome.
- T-test compares the statistical difference of two groups (**binary class**) and used for **continuous** features
- ANOVA uses variance to test the relationship between **categorical** predictors and **continuous** outcome response (e.g. gender, age group to predict exam mark).
- Correlation test work for predictors and outcome are both **continuous**.
- Assume a **null-hypothesis**. Use p value to **reject** the null-hypothesis (e.g. **$p < 0.01$**). Smaller p value indicates the evidence against the null-hypothesis.



Questions

Name a nonlinear dimensionality reduction technique?



Popular Dimensionality Reduction Methods

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Manifold learning (non-linear)



Difference between multinomial naïve bayes and Gaussian naïve bayes?



Naïve Bayes

- Based on Bayes' theorem $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- Find the probability of **A** happening, given that **B** has occurred.
- Assumption is that features are independent with the same importance (Naïve).
- Naïve Bayes leads to a linear decision boundary
- Multinomial Naïve Bayes (features are counts: number of occurrence)
- Gaussian Naïve Bayes (features are continuous values)
- Bernoulli Naïve Bayes (features are only binary values: yes or no)



Multinomial Naïve Bayes

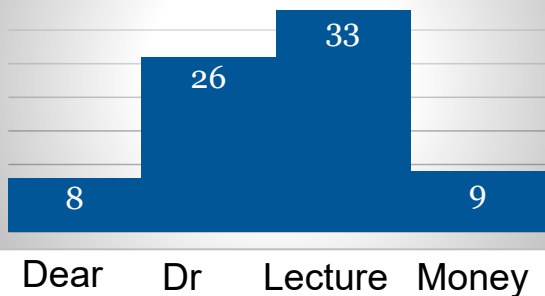
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Application: Spam email detection

Test: **Lecture, Money**

Training Data

Normal



$P(B|A)$

$$\begin{aligned} P(\text{Dear}|N) &= 8/76 = 0.105 \\ P(\text{Dr}|N) &= 26/76 = 0.342 \\ P(\text{Lecture}|N) &= 33/76 = 0.434 \\ P(\text{Money}|N) &= 9/76 = 0.118 \end{aligned}$$

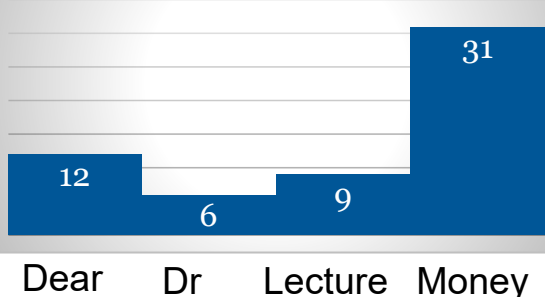
$P(A)$: Prior probability

$$P(N) = 76/(76+58) = 0.57$$

$$P(A|B) \propto P(B|A)P(A)$$

$$\begin{aligned} P(N | \text{Lecture, Money}) &= P(N) * P(\text{Lecture}|N) * P(\text{Money}|N) \\ &= 0.57 * 0.434 * 0.118 \\ &= 0.029 \end{aligned}$$

Spam



$$\begin{aligned} P(\text{Dear}|S) &= 12/58 = 0.207 \\ P(\text{Dr}|S) &= 6/58 = 0.103 \\ P(\text{Lecture}|S) &= 9/58 = 0.155 \\ P(\text{Money}|S) &= 31/58 = 0.534 \end{aligned}$$

$$P(S) = 58/(76+58) = 0.43$$

$$\begin{aligned} P(S | \text{Lecture, Money}) &= P(S) * P(\text{Lecture}|S) * P(\text{Money}|S) \\ &= 0.43 * 0.155 * 0.534 \\ &= 0.036 \end{aligned}$$

0.036 > 0.029, it is a Spam!



Gaussian Naïve Bayes

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

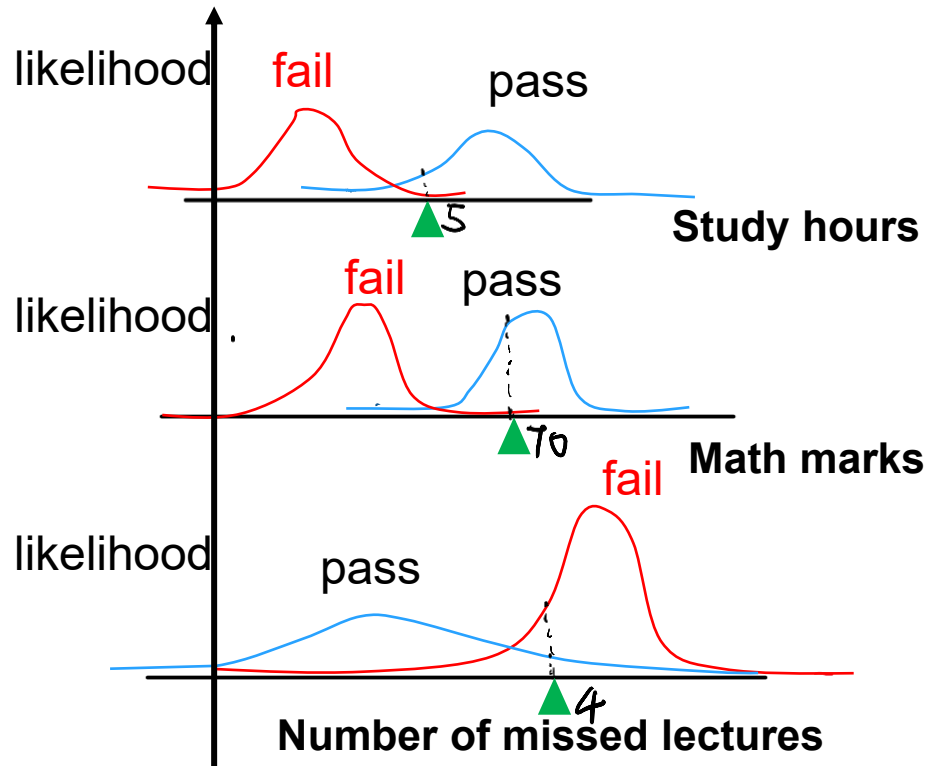
Application: if you'll pass Machine Learning

Training Data: convert data to Gaussian distributions

$P(A)$: Prior probability

Test: study hours (5), math marks (70), missed(4)

$$P(A|B) \propto P(B|A)P(A)$$



$$P(P) = 0.7$$

$$\begin{aligned} P(P|\text{hours}=5, \text{mark}=70, \text{missed}=4) \\ &= P(P) * L(\text{hours}=5|P) * L(\text{mark}=70|P) * L(\text{missed}=4|P) \\ &= 0.7 * 0.03 * 0.08 * 0.01 \end{aligned}$$

We use **log** to avoid underflow:

$$\begin{aligned} &\log(0.7 * 0.03 * 0.08 * 0.01) \\ &= \log(0.7) + \log(0.03) + \log(0.08) + \log(0.01) \\ &= -10.99 \end{aligned}$$

$$P(F) = 0.3$$

$$\begin{aligned} P(F|\text{hours}=5, \text{mark}=70, \text{missed}=4) \\ &= P(F) * L(\text{hours}=5|F) * L(\text{mark}=70|F) * L(\text{missed}=4|F) \\ &= 0.3 * 0.00001 * 0.000001 * 0.04 \end{aligned}$$

$$\begin{aligned} &\log(0.3) + \log(0.00001) + \log(0.000001) + \log(0.04) \\ &= -29.75 \end{aligned}$$

-10.99 > -29.75, it is a Pass!



Questions

What is the hyperparameter in KNN?

K Nearest Neighbour (KNN)

Input: the training set T , category label set C and test object x . Each object has features x_1 and x_2 . The number of neighbours K .

Output: the category c_x of the test object x (c_x belongs to C)

begin

for each t belongs to T *do*

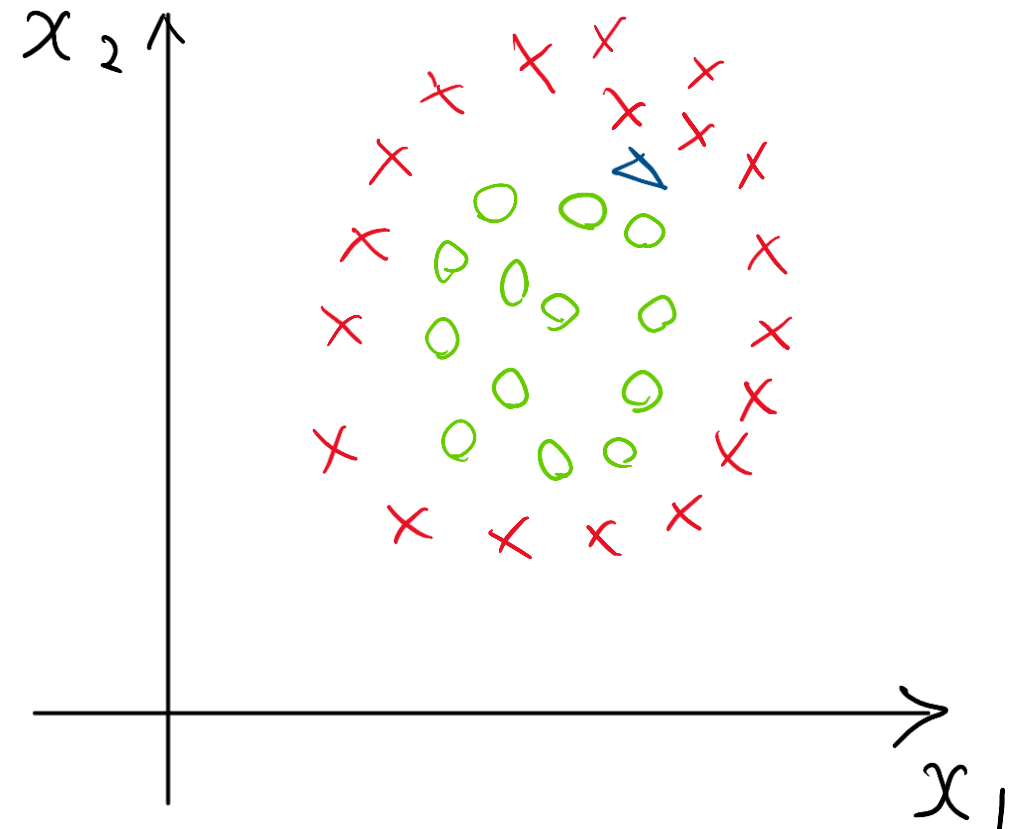
 calculate the distance $D(t, x)$

end for

 select K training examples from T that have the smallest distance D .

 calculate c_x as the majority class in the K examples.

end

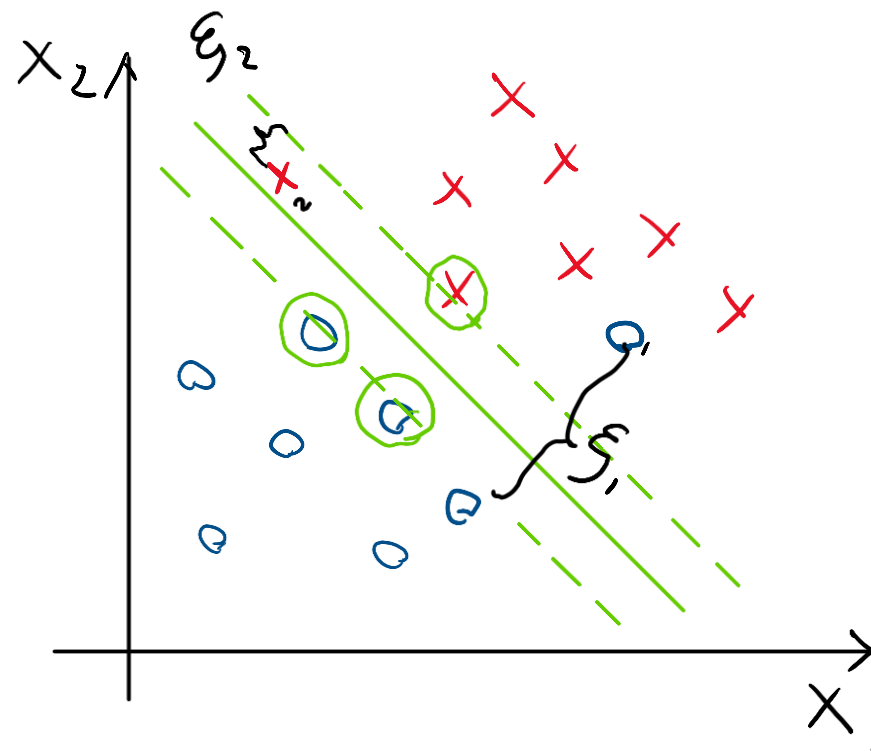
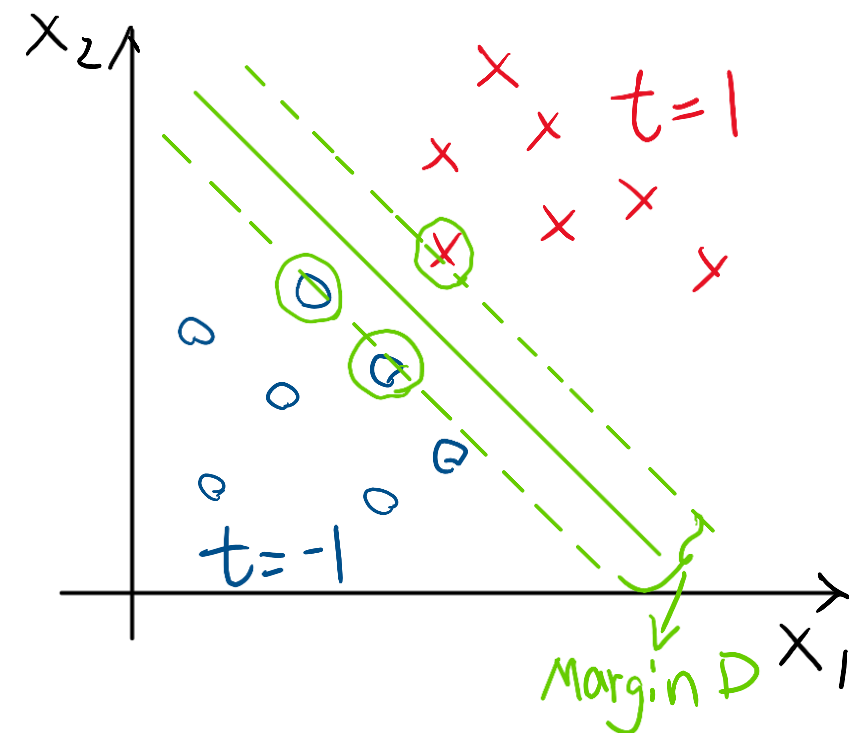




Differences between hard margin SVM and soft margin SVM?



$n=(1, \dots, N)$ data points





Random Forest

Why random forest is called “random”?



Random Forest

- **N samples (e.g. 1000), F features (e.g.100), B trees (e.g. 500 trees):**
 - Final decision is based on majority voting from B decisions.
 - Regression is based on average of B trees.
 - Uncertainty can be obtained based on ratios of positive/negative votes from B trees.
- **Bagging:** the ensemble learning method that is commonly used to reduce **variance** within a noisy dataset.
 - For each of B trees, use randomly sampled 80% data for training and 20% for validation
- **Random Subspaces:** randomly sampling the features (only use a random subset of features for each tree)
 - Don't always rely on the same set of features.
 - To avoid certain features overly emphasised.
 - Classification: $F^{0.5}$, Regression: $F/3$.



Metrics Recap

A model outputs 5 TP, 6 TN, 3 FP, and 2 FN. Calculate the recall.

0.714

☐

0.625

☐

0.455

☐



You're building a binary classifier that checks photos of insect traps for whether a dangerous invasive species is present. If the model detects the species, the entomologist (insect scientist) on duty is notified. Early detection of this insect is critical to preventing an infestation. A false alarm (false positive) is easy to handle: the entomologist sees that the photo was misclassified and marks it as such. Assuming an acceptable accuracy level, which metric should this model be optimized for?

False positive rate (FPR)

☐

Precision

☐

Recall

☐



Metrics Recap

A model outputs 3 TP, 4 TN, 2 FP, and 1 FN. Calculate the precision.

0.75

☐

0.6

☐

0.429

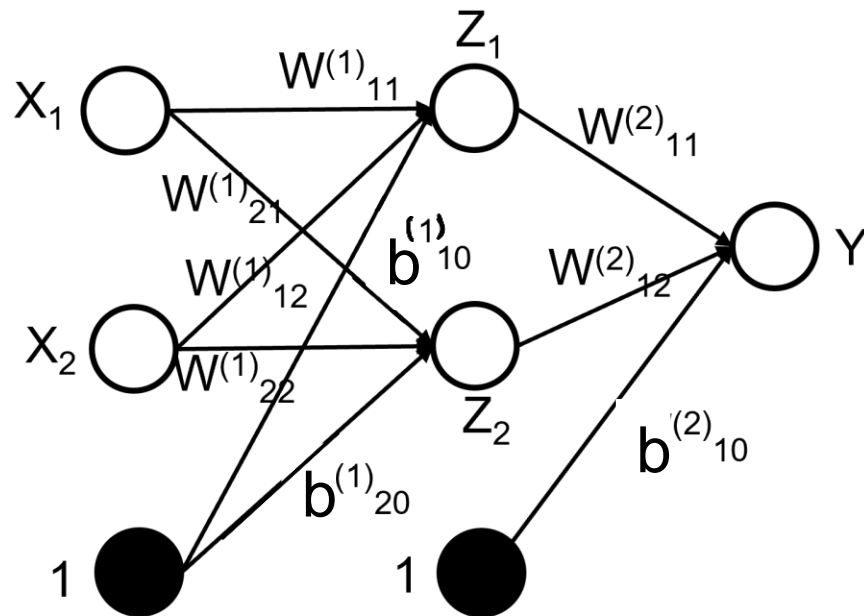
☐

Optimisation (Back Propagation)

Example:

Activation function at Z : $f_z(a) = a^2 + 3a$

Activation function at Y : $f_y(a) = a$



Loss function MSE: $L = \frac{1}{2N} \sum_{n=1}^N (Y_n - \hat{Y}_n)^2$

Use gradient descent to update w and b .

How to update $w^{(2)}_{11}$?

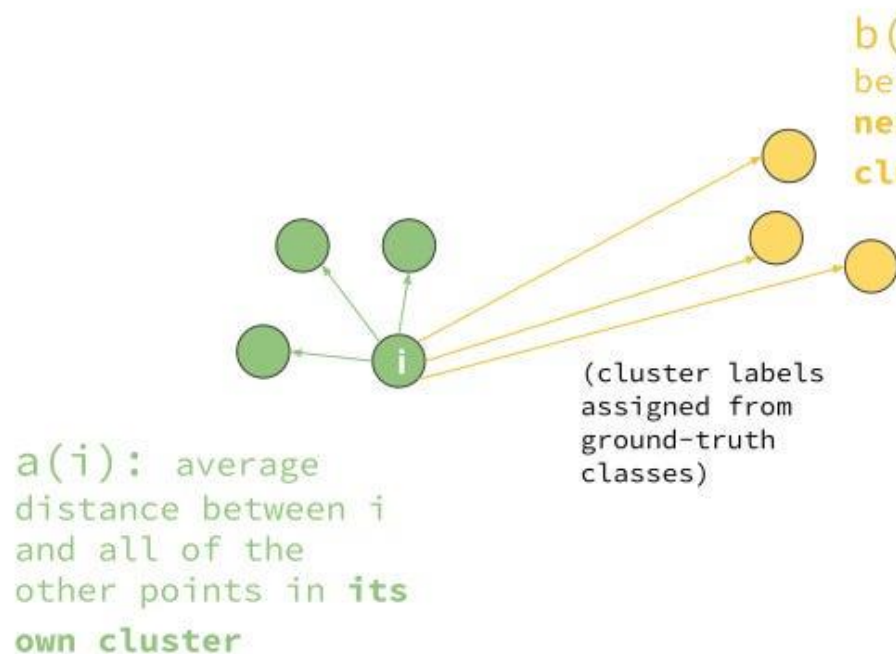
$$\frac{\partial L}{\partial w^{(2)}_{11}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial a} \frac{\partial a}{\partial w^{(2)}_{11}}$$

$$\frac{\partial L}{\partial \hat{y}} = -\frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n), \quad \frac{\partial \hat{y}}{\partial a} = 1$$

$$\frac{\partial a}{\partial w^{(2)}_{11}} = \frac{\partial (z_1 w^{(2)}_{11} + z_2 w^{(2)}_{12} + b^{(2)}_{10})}{\partial w^{(2)}_{11}} = z_1$$

$$\frac{\partial L}{\partial w^{(2)}_{11}} = -\frac{1}{N} \sum_{n=1}^N (y_n - \hat{y}_n) z_1, \quad w^{(2)'}_{11} = w^{(2)}_{11} - \alpha \frac{\partial L}{\partial w^{(2)}_{11}}$$

Evaluation of Cluster Analysis - SC



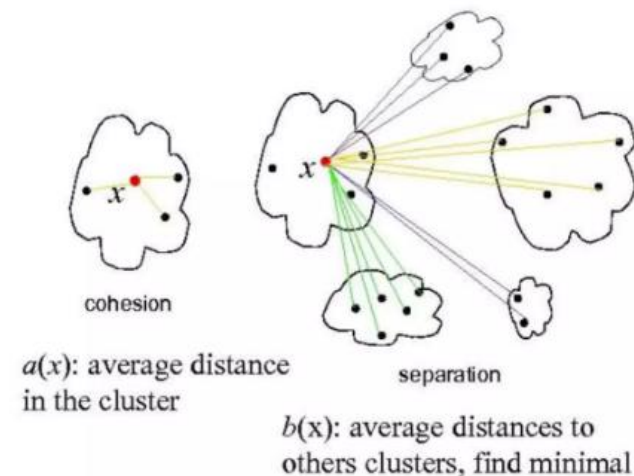
For a single point, i

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

$-1 \leq s(i) \leq 1$

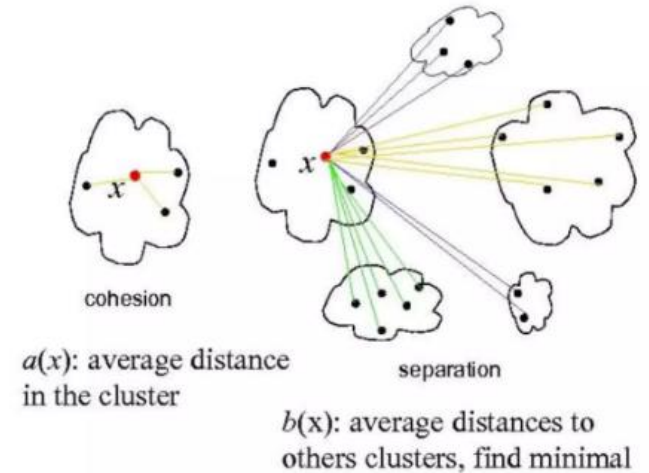
Sprawling, overlapped clusters

Tight, well-separated clusters



Evaluation of Cluster Analysis - SC

1. For each data point, calculate two values:
 - *Average distance to all other data points within the same cluster (cohesion).*
 - *Average distance to all data points in the nearest neighboring cluster (separation).*
2. Compute the silhouette coefficient for each data point using the formula:
$$\text{silhouette coefficient} = (\text{separation} - \text{cohesion}) / \max(\text{separation}, \text{cohesion})$$
3. Calculate the average silhouette coefficient across all data points to obtain the overall silhouette score for the clustering result.



SC Example

Cluster 1:

Point A1: (2, 5)

Point A2: (3, 4)

Point A3: (4, 6)

Step 1: Calculate Cohesion for Point A1

•Distance from Point A1 to Point A2: $\sqrt{(2-3)^2 + (5-4)^2} =$

$\sqrt{2}$

•Distance from Point A1 to Point A3: $\sqrt{(2-4)^2 + (5-6)^2} =$

$\sqrt{5}$

•Average cohesion for Point A1 = $(\sqrt{2} + \sqrt{5}) / 2 \approx 1.825$

Cluster 2:

Point B1: (8, 3)

Point B2: (9, 2)

Point B3: (10, 5)

Step 2: Calculate Separation for Point A1

•Distance from Point A1 to Point C1: $\sqrt{(2-6)^2 + (5-10)^2} =$

$\sqrt{26}$

•Distance from Point A1 to Point C2: $\sqrt{(2-7)^2 + (5-8)^2} =$

$\sqrt{18}$

•Distance from Point A1 to Point C3: $\sqrt{(2-8)^2 + (5-9)^2} =$

$\sqrt{32}$

•Average separation for Point A1 = $(\sqrt{26} + \sqrt{18} +$

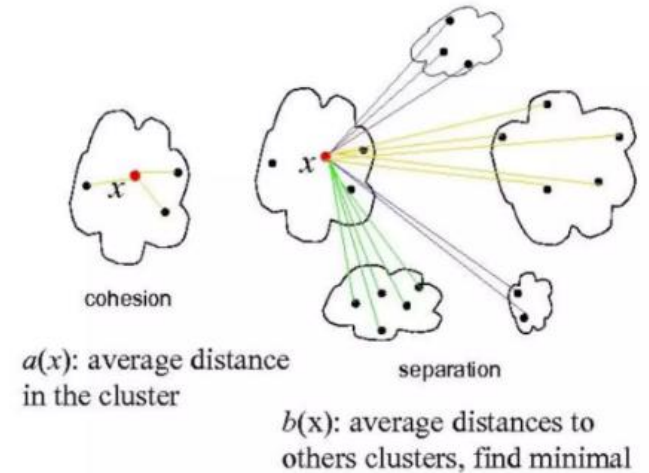
$\sqrt{32}) / 3 \approx 5.333$

Cluster 3:

Point C1: (6, 10)

Point C2: (7, 8)

Point C3: (8, 9)



SC Example

Cluster 1:

Point A1: (2, 5)

Point A2: (3, 4)

Point A3: (4, 6)

Step 3: Calculate Silhouette Coefficient for Point A1

•silhouette coefficient = (separation — cohesion) /
max(separation, cohesion)

•silhouette coefficient = $(5.333 - 1.825) / \max(5.333, 1.825) \approx 0.657$

Cluster 2:

Point B1: (8, 3)

Point B2: (9, 2)

Point B3: (10, 5)

Step 4: Average Silhouette Coefficient

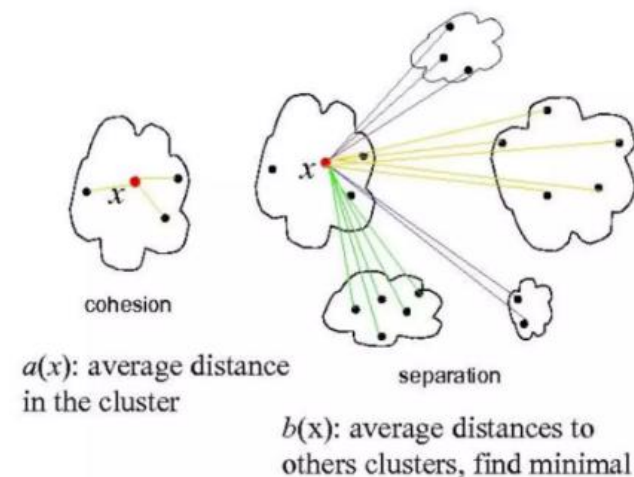
Calculate the average silhouette coefficient across all data points to obtain the overall silhouette score for the clustering result.

Cluster 3:

Point C1: (6, 10)

Point C2: (7, 8)

Point C3: (8, 9)

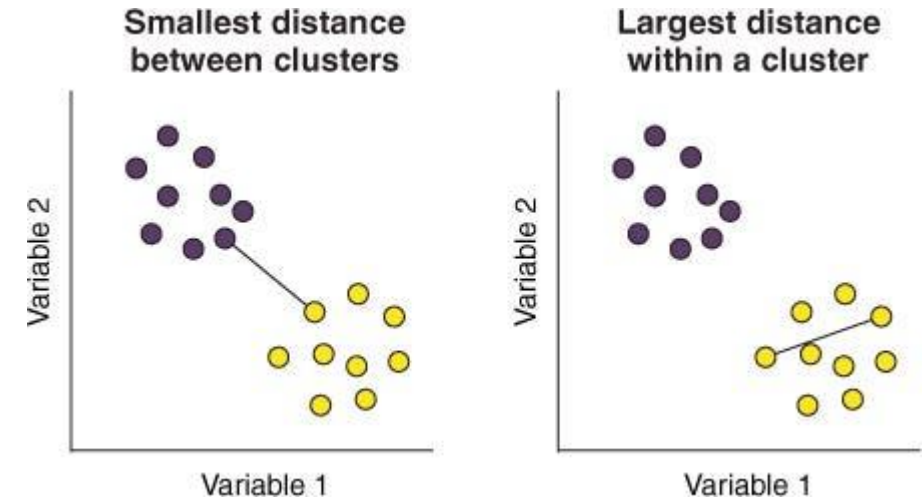


Evaluation of Cluster Analysis - DI

- Dunn's Index:

$d(i,j)$ represents the **inter-cluster** distance (centroids between two clusters) between i and j .
 $d'(k)$ measures the **intra-cluster** distance of cluster k .

$$DI = \frac{\min_{1 \leq i < j \leq n} d(i,j)}{\max_{1 \leq k \leq n} d'(k)}$$



Evaluation of Cluster Analysis - DI

Intra-cluster distance

It refers to the average distance between data points within the same cluster. In other words, it measures the compactness or cohesion of data points within a cluster. The smaller the intra-cluster distance, the more similar and tightly packed the data points are within the cluster.

Distance between A and B:

$$(x_A - x_B)^2 + (y_A - y_B)^2$$

$$(2 - 3)^2 + (4 - 5)^2$$

$$1^2 + (-1)^2$$

$$1 + 1$$

$$2$$

A: (2, 4)

B: (3, 5)

C: (5, 7)

Distance between A and C:

$$(x_A - x_C)^2 + (y_A - y_C)^2$$

$$(2 - 5)^2 + (4 - 7)^2$$

$$(-3)^2 + (-3)^2$$

$$9 + 9$$

$$18$$

Distance between B and C:

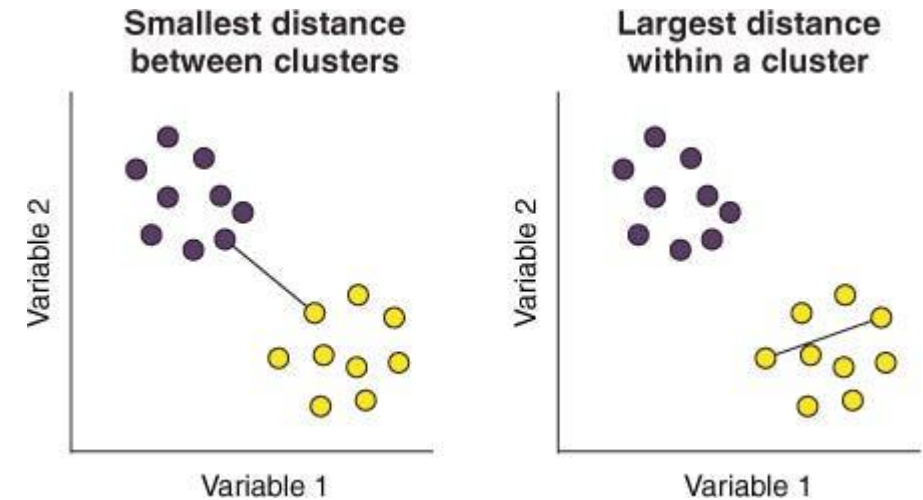
$$(x_B - x_C)^2 + (y_B - y_C)^2$$

$$(3 - 5)^2 + (5 - 7)^2$$

$$(-2)^2 + (-2)^2$$

$$4 + 4$$

$$8$$



$$\text{Average distance} = (2 + 18 + 8) / 3 = 28 / 3 \approx 9.333$$

$$\text{Maximum distance} = \max(2, 18, 8) = 18$$

Evaluation of Cluster Analysis - DI

Inter-cluster distance

It refers to the average distance between different clusters in a clustering solution. It measures the separation or dissimilarity between clusters. Inter-cluster distance is usually calculated as the distance between the centroids (mean or center points) of the clusters or as the minimum distance between data points in different clusters.

Distance between Cluster 1 and Cluster 2:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{(6 - 2)^2 + (8 - 4)^2}$$

$$\sqrt{4^2 + 4^2}$$

$$\sqrt{16 + 16}$$

$$\sqrt{32}$$

Inter-cluster distance between Cluster 1 and

Cluster 2 ≈ 5.657

Distance between Cluster 1 and Cluster 3:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{(10 - 2)^2 + (12 - 4)^2}$$

$$\sqrt{8^2 + 8^2}$$

$$\sqrt{64 + 64}$$

$$\sqrt{128}$$

Inter-cluster distance between Cluster 1 and

Cluster 3 ≈ 11.314

Distance between Cluster 2 and Cluster 3:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

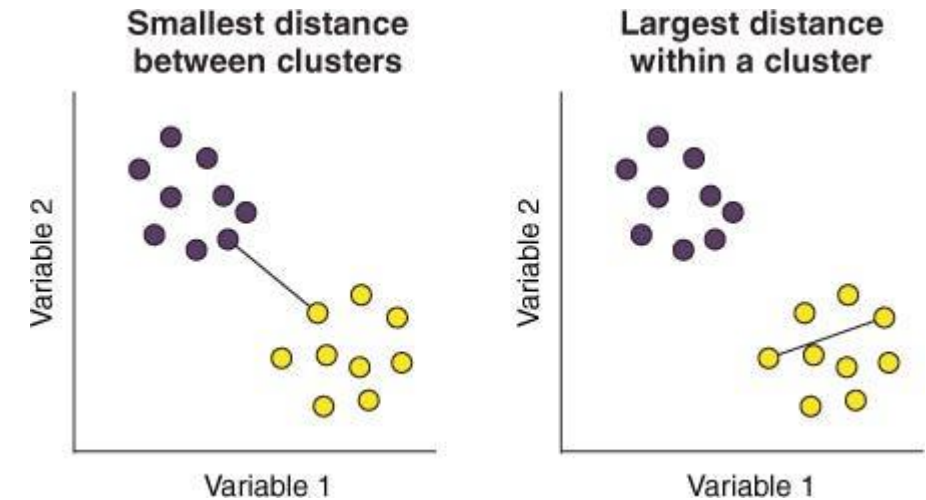
$$\sqrt{(10 - 6)^2 + (12 - 8)^2}$$

$$\sqrt{4^2 + 4^2}$$

$$\sqrt{16 + 16}$$

$$\sqrt{32}$$

Inter-cluster distance between Cluster 2
and Cluster 3 ≈ 5.657



$$DI = \frac{\min_{1 \leq i < j \leq n} d(i, j)}{\max_{1 \leq k \leq n} d'(k)}$$

Inter-cluster distance between Cluster 1 and Cluster 2 ≈ 5.657

Inter-cluster distance between Cluster 1 and Cluster 3 ≈ 11.314

Inter-cluster distance between Cluster 2 and Cluster 3 ≈ 5.657

The minimum distance among all the pairs is 5.657



NN Questions

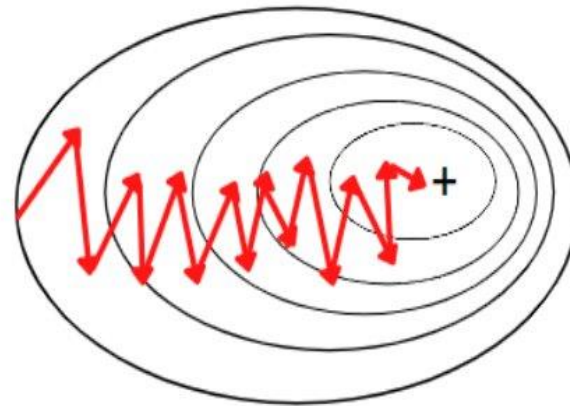
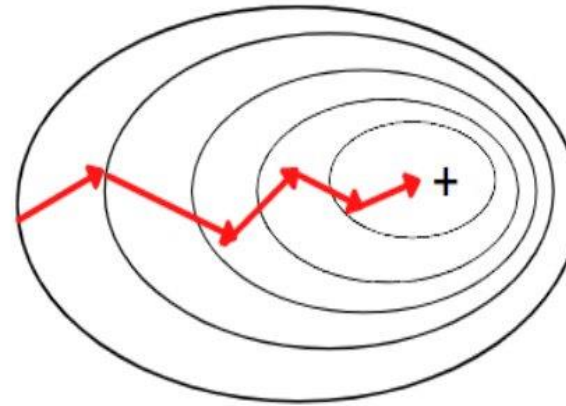
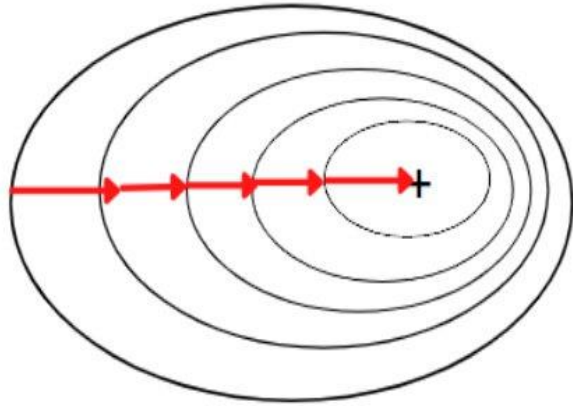
1. Fill in the blank in the following sentence:

Neural network models can automatically learn ____ during training.

- ☐ Activation functions
- ☐ Hyperparameters
- ☐ Nonlinear feature interactions
- ☐ The loss function



Gradient Descent



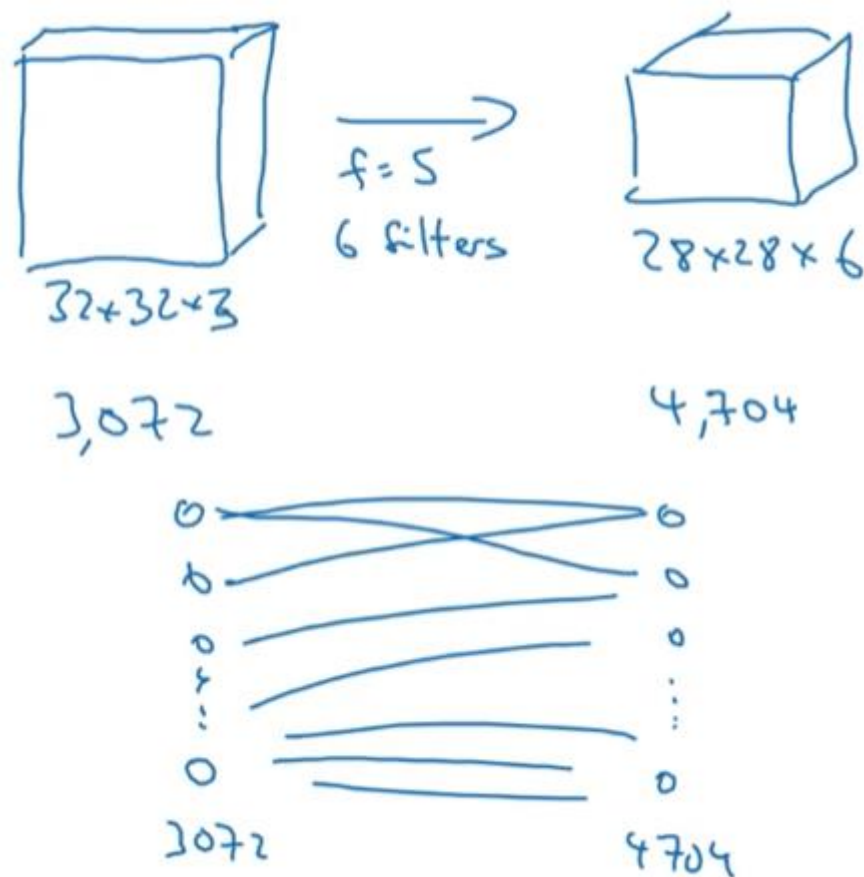


Number of parameters in the CNN

	Activation shape	Activation Size	# parameters
Input:	(32,32,3)	3,072 $a^{[0]}$	0
CONV1 (f=5, s=1)	(28,28,8)	6,272	
POOL1	(14,14,8)	1,568	
CONV2 (f=5, s=1)	(10,10,16)	1,600	
POOL2	(5,5,16)	400	
FC3	(120,1)	120	
FC4	(84,1)	84	
Softmax	(10,1)	10	

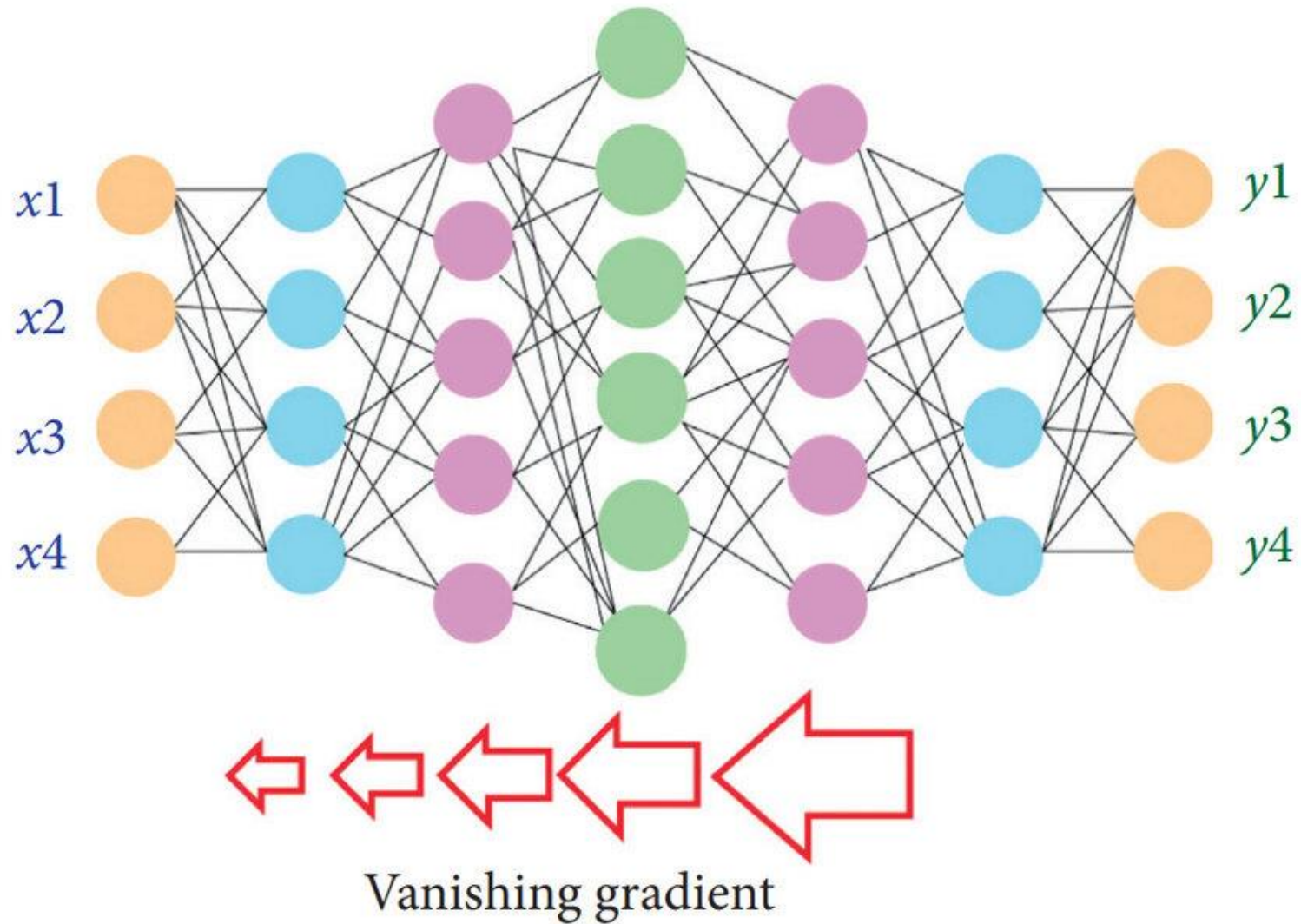


Comparing CNN vs ANN



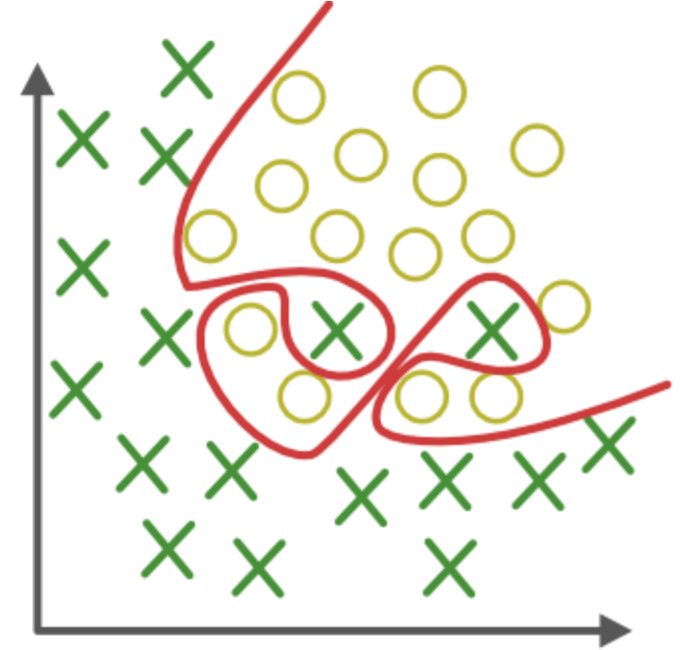
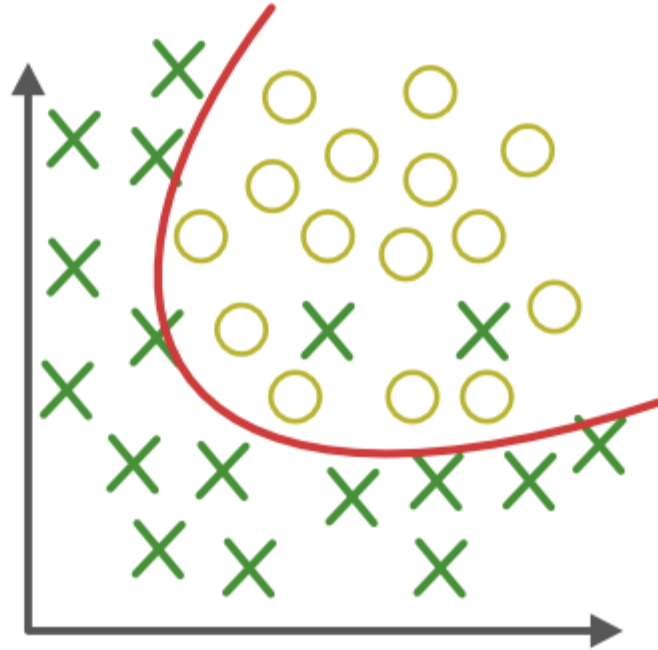
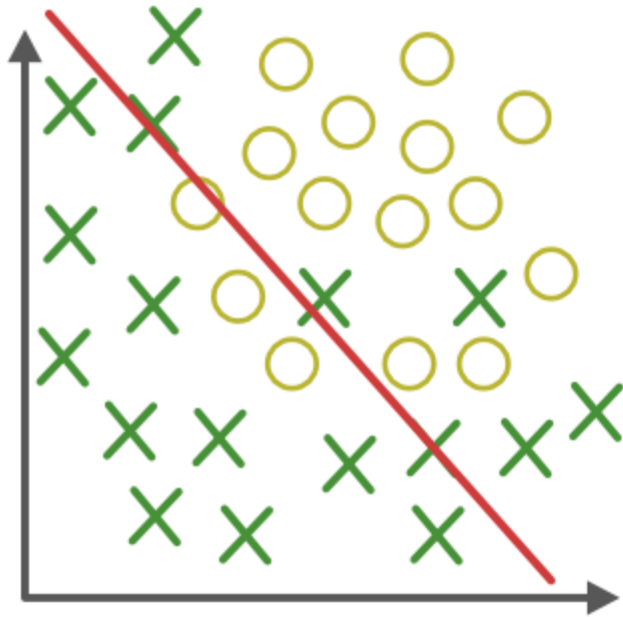


Vanishing Gradient





Overfitting



Solutions?